Guide to Interpreting the Output of the Segmented Fourier Analysis Code (Version 9.4)

1 Overview

The v. 9.4 code takes as input a list of pairs (n, μ_n), where μ_n is the minimum modulus among the eigenvalues of a certain matrix associated to the index n. Its purpose is to determine whether the sequence $\{\mu_n\}$ oscillates in a nearly periodic fashion and, if so, to estimate the period.

What changed from earlier versions

Versions 8 and 9 attempted to remove any overall trend (growth, decay, curvature) from the data by fitting a single curve to the entire dataset before analyzing the residual for periodicity. This failed on data composed of qualitatively different regions—for instance, a linearly rising segment followed by a flat plateau. No single polynomial or smoother can follow the sharp bend between such regions, and the resulting artifact corrupts all downstream analysis.

Version 9.4 takes a fundamentally different approach:

1. **Segmentation.** Automatically detect *breakpoints* where the data's behavior changes, and split the data into segments at those points.

2. **Per-segment analysis.** Within each segment, the data is approximately linear, so a simple straight-line fit suffices to remove the trend. Each segment is then analyzed independently for periodicity.

The report therefore contains one verdict per segment rather than a single verdict for the whole dataset.

2 Glossary of abbreviations and acronyms

ACF — **Autocorrelation function.** Given a sequence $x_1, \ldots, x_n$ with mean $\bar{x}$, the normalized autocorrelation at lag k is

$$\mathrm{ACF}(k) = \frac{\sum_{i=1}^{n-k}(x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

By construction $\mathrm{ACF}(0) = 1$. If the sequence repeats with period T, the ACF has a local maximum near 1 at lag T. See any introductory time series text, or the Wikipedia article *Autocorrelation*, for further background.

ACF strength — The value of $\mathrm{ACF}(T)$ at the detected period T. Ranges from -1 to $+1$. Values near $+1$ indicate strong, clean repetition. The code uses 0.5 as the threshold for a confident detection and 0.2 as the threshold below which the detection is dismissed.

AR(1) **First-order autoregressive model.** A stochastic process in which each value depends linearly on its predecessor:

$$x_n = \rho\, x_{n-1} + \varepsilon_n, \qquad \varepsilon_n \sim \mathcal{N}(0, \sigma^2).$$

The parameter ρ ("rho") is the lag-1 autocorrelation. This model serves as the *null hypothesis*: it represents data that has short-range correlation but no true periodicity. A spectral peak is declared significant only if it exceeds what the AR(1) model would produce. See Gilman, Fuglister, and Mitchell, "On the power spectrum of 'red noise'" *J. Atmos. Sci.* **20** (1963), 182–184, for the theoretical power spectrum used.

Bonferroni correction

A multiple-testing correction. When m frequency bins are tested simultaneously at significance level α, up to $m\alpha$ false positives are expected by chance. The Bonferroni correction tests each bin at level α/m, so that the probability of *any* false positive across all bins stays at α. Conservative but simple and widely accepted.

Breakpoint An index n at which the data's local behavior changes qualitatively—for example, from rising to flat, or from one slope to another. Detected by the PELT algorithm (see below).

dB **Decibel.** $10 \log_{10}(\text{power})$. A logarithmic unit for power, used in some plots.

Dynamic range

The ratio of the 90th percentile to the 10th percentile of the estimated envelope. If this exceeds 3, the code concludes that the oscillation amplitude varies enough to require normalization.

Envelope The slowly varying curve tracing the peaks of an oscillating signal. For $A(n)\sin(2\pi n/T)$, the envelope is $|A(n)|$.

FFT **Fast Fourier Transform.** An $O(n \log n)$ algorithm for computing the discrete Fourier transform of a length-n sequence, decomposing it into sinusoidal components at frequencies k/n, $k = 0, 1, \ldots, n-1$. See any numerical analysis text or the Wikipedia article *Fast Fourier transform*.

Fundamental period

The true repeat period of the data, as distinguished from harmonics (integer fractions of the period) and integer multiples (longer lags at which the ACF also peaks). Identified by the multi-scale ACF procedure described in Section 5.5.

Hann window A bell-shaped taper $w_k = \frac{1}{2}\big(1 - \cos(2\pi k/(n-1))\big)$ applied to the data before the FFT. It reduces *spectral leakage* (spurious spreading of power across frequencies caused by the finite length of the data) but attenuates signals near the Nyquist frequency.

Harmonic An integer multiple of a fundamental frequency. If the fundamental frequency is $f_0 = 1/T$, the m-th harmonic is mf_0, corresponding to period T/m. Non-sinusoidal waveforms (e.g., sawtooth, square wave) have energy at many harmonics; the FFT reports these as separate peaks.

MA(w) **Moving average with window width w.** The value at index i is replaced by the mean of the w values centered at i. Used in the multi-scale period detection to suppress fine-scale structure.

Nyquist frequency The highest frequency representable by a discrete sequence: $f_{\text{Nyq}} = 1/2$ (cycles per sample), corresponding to a period of 2. The Hann window attenuates signals near this limit, which is why a period-2 oscillation can be missed by the FFT while the ACF detects it.

$n/3$ The maximum period considered credible within a segment of length n. A period of T in n points gives n/T full cycles; for $T > n/3$ there are fewer than three full cycles, which is insufficient to distinguish periodicity from a trend artifact.

PELT **Pruned Exact Linear Time.** A changepoint detection algorithm that finds the positions and number of breakpoints in a sequence by minimizing a penalized cost function. The penalty (proportional to $\log(N) \cdot \text{Var}(\mu)$ in this code) controls sensitivity: a larger penalty produces fewer breakpoints. Reference: Killick, Fearnhead, and Eckley, "Optimal detection of changepoints with a linear computational cost," *J. Amer. Statist. Assoc.* **107** (2012), 1590–1598. The implementation used is from the `ruptures` Python library (Truong, Oudre, and Vayatis, 2020).

Power The squared magnitude $|c_k|^2$ of a Fourier coefficient, measuring how much of the signal's variance is attributable to oscillation at frequency k/n. Displayed in scientific notation: `1.58e+04` means 1.58×10^4.

Power spectrum The function mapping each frequency to its power. A peak in the power spectrum indicates energy concentrated at that frequency.

rho (ρ) The lag-1 autocorrelation of the analysis signal, used as the AR(1) parameter. ρ near $+1$: strong positive correlation. ρ near -1: alternation. ρ near 0: near-independence.

Savitzky–Golay filter A smoothing method that fits a low-degree polynomial by least squares within a sliding window, replacing each value with the fitted value. Used here for envelope estimation. Reference: Savitzky and Golay, *Anal. Chem.* **36** (1964), 1627–1639.

Spectral leakage Spurious spreading of a signal's power across neighboring frequencies, caused by the FFT's implicit assumption that the data is periodic over its length. Reduced by applying a Hann window before the FFT.

std dev Standard deviation.

3 Section 1 of the report: data summary

The report begins with the range of the index n, the number of data points N, and the range of the minimum moduli μ_n.

4 Section 2 of the report: segmentation

The code approximates the data by a piecewise-linear function—that is, a sequence of straight line segments joined at breakpoints—and finds the breakpoint positions that minimize the total squared error, subject to a penalty for each additional breakpoint. This is done by the PELT algorithm (see glossary).

The report lists:

- the number of breakpoints detected,
- for each resulting segment: the index range, the number of points, and the range of μ_n values within that segment.

Example. Data that rises linearly from $n = 1$ to $n = 120$ and then levels off would produce one breakpoint near $n = 120$, giving two segments: the ramp (Segment 1) and the plateau (Segment 2).

If no breakpoints are detected, the entire dataset is treated as a single segment.

5 Section 3 of the report: per-segment analysis

Each segment is analyzed independently. The analysis has five stages, described in the subsections below. Segments shorter than 10 points are skipped.

5.1 Linear detrend

A straight line is fitted to the segment by least squares and subtracted. Within a single segment, the data is approximately linear (that is the whole point of segmenting), so this simple detrend is adequate. The report prints the slope and the standard deviation of the residual.

5.2 Envelope normalization

If the oscillation amplitude varies strongly within a segment (for instance, an oscillation that decays as $1/n$), the ACF and significance tests become unreliable. The code estimates the local amplitude envelope by smoothing the absolute values of the detrended signal with a Savitzky–Golay filter, then checks the *dynamic range* (90th percentile divided by 10th percentile of the envelope). If the dynamic range exceeds 3, the code divides the detrended signal by the envelope, producing a signal with roughly constant amplitude.

When normalization is applied, the report notes this together with the dynamic range. When it is not needed, the report says so. All subsequent analysis operates on the signal after this step, called the *analysis signal*.

5.3 Windowed FFT

The analysis signal is multiplied by a Hann window and passed through the FFT, yielding a power spectrum. See the glossary entries for FFT, Hann window, power spectrum, and spectral leakage.

5.4 Significance testing

To determine whether a peak in the power spectrum represents genuine periodicity rather than noise, the code constructs an AR(1) null model (see glossary) fitted to the analysis signal. The theoretical power spectrum of this null model, which is not flat but tilted toward low or high frequencies depending on the sign of ρ, serves as the baseline.

The 95% and 99% confidence thresholds are computed from a chi-squared distribution with 2 degrees of freedom (the approximate distribution of FFT power under the null), then inflated by the Bonferroni correction (see glossary) to account for the number of frequency bins tested.

Peaks exceeding the threshold are listed in a table with their period, power, and significance level (95% or 99%).

5.5 Multi-scale ACF period detection

The FFT can misidentify the fundamental period of non-sinusoidal waveforms, because sharp features (sawtooth teeth, square-wave edges) distribute energy across many harmonics, and a harmonic can exceed the fundamental in power. The multi-scale ACF procedure avoids this by working directly with the autocorrelation, which is insensitive to waveform shape.

The procedure examines the analysis signal at progressively coarser scales by smoothing it with moving averages of increasing width. At each scale, it computes the ACF and records the first peak whose strength exceeds 0.1. Fine scales detect short-period features; coarse scales detect longer periods after the short-period structure has been averaged out.

Identifying the fundamental

Among all detected periods, the code selects the one whose integer multiples best account for the others. For each candidate period T_c, it counts how many of the other detected periods are approximately equal to mT_c for some positive integer m (within a 15% tolerance). The candidate explaining the most others wins; ties are broken by ACF strength.

This correctly handles two situations:

- A period-2 oscillation whose ACF peaks at lags $2, 4, 6, 8$: period 2 explains all the others as multiples, so it is the fundamental.
- A period-60 sawtooth with sub-cycle teeth of period 3: period 60 and period 3 may both appear, but the tie is broken by ACF strength, which favors the coarser-scale detection of period 60.

6 Verdicts

Each segment receives one of four verdicts:

PERIODIC At least one of the following holds: (a) the FFT has a peak exceeding the Bonferroni-corrected 95% or 99% threshold, or (b) the multi-scale ACF finds a fundamental with strength above 0.5. The fundamental period and ACF strength are reported.

PERIODIC (FFT only)
The FFT has significant peaks, but the multi-scale ACF does not find a strong fundamental. The dominant FFT period is reported.

POSSIBLY PERIODIC
The multi-scale ACF finds a candidate period with strength between 0.2 and 0.5, but the FFT does not confirm. The candidate period is reported with a caution.

NOT PERIODIC

Neither the FFT nor the ACF finds evidence of oscillatory behavior in this segment.

7 Overall summary

After all segments have been analyzed, the report prints a one-line summary for each segment: the index range, the verdict, and (if applicable) the fundamental period and ACF strength. This is the primary output for quick reference.

8 Visualizations

8.1 Overview plots (one figure)

1. **Original Data with Detected Breakpoints.** The full dataset with vertical red dashed lines at each breakpoint. Verify that the breakpoints fall where the data's character visibly changes.

2. **Segment-wise Detrended Residuals.** The residuals from each segment's linear detrend, plotted in different colors. Each segment's residual should oscillate around zero with no visible drift. If a segment's residual has a large broad hump or slope, the segmentation may have missed a breakpoint.

8.2 Per-segment detail plots (one figure per segment)

Each segment produces four panels:

1. **Data with Linear Trend.** The segment's data (blue) with the fitted line (red). The line should follow the broad slope of the data without tracking oscillations.

2. **Analysis Signal.** The signal that the FFT and ACF operate on (after detrending and, if applicable, envelope normalization). It should oscillate around zero with roughly constant amplitude.

3. **Power Spectrum.** Log-scale plot of power versus frequency. The dashed lines are the AR(1) null model (black), the 95% Bonferroni-corrected threshold (green), and the 99% threshold (red). Colored dots mark peaks exceeding a threshold.

4. **Power vs Period.** The same information as the power spectrum, but with the horizontal axis showing period (the reciprocal of frequency) rather than frequency. Restricted to periods between 2 and $n/3$.

9 Practical notes

When the segmentation is wrong

The PELT algorithm's penalty parameter controls sensitivity. The code sets it to $\log(N) \cdot \mathrm{Var}(\mu)$, which is a standard default. If you believe a breakpoint was missed or spuriously inserted, the penalty can be adjusted by editing the line `penalty_value = np.log(N) * np.var(minmoduli)` in the code. Increasing the penalty produces fewer breakpoints; decreasing it produces more.

When a segment has no breakpoints

If the data is already well-described by a single straight line (or is roughly flat with oscillations), no breakpoints will be detected, and the entire dataset is analyzed as one segment. The analysis within that segment is identical to what earlier versions did (minus the complications of global polynomial or Savitzky–Golay detrending).

Dependencies

Version 9.4 requires the `ruptures` Python library, which is not included in SageMath by default. Install it with `!pip install ruptures` in a SageMath notebook cell, or `pip install ruptures` from the command line.